Amazon Auto Scaling — Important Interview Questions & Answers

1.What is Auto Scaling?
Auto Scaling is an AWS service that automatically adjusts the number of EC2 instances based on application demand. It helps maintain application availability and can also help control costs.
Simple ga gurthupettuko:
Auto Scaling ante application demand batti EC2 instances ni automatic ga increase or decrease chese service.
Eng: Auto Scaling automatically adjusts the number of EC2 instances based on demand.

2.What is an Auto Scaling Group?
An Auto Scaling Group is a collection of EC2 instances that are managed together by Auto Scaling. It maintains the required number of instances and replaces unhealthy instances when needed.
Simple ga gurthupettuko:
Auto Scaling Group ante EC2 instances ni group ga manage chese service.
Eng: An Auto Scaling Group manages a group of EC2 instances.

3.What is a Launch Template?
A Launch Template contains the configuration required to launch EC2 instances. It can include the AMI, instance type, key pair, security groups, storage, and other settings.
Simple ga gurthupettuko:
Launch Template ante EC2 create cheyyadaniki required configuration store chese template.
Eng: A Launch Template contains the configuration required to launch EC2 instances.

4.What is a Scaling Policy?
A Scaling Policy defines when and how Auto Scaling should increase or decrease the number of EC2 instances. It is used based on application demand or specific conditions.
Simple ga gurthupettuko:
Scaling Policy ante EC2 instances eppudu increase or decrease cheyyalo define chese rule.
Eng: A Scaling Policy defines when and how Auto Scaling should scale instances.

5.What is Dynamic Scaling?
Dynamic Scaling automatically adjusts the number of EC2 instances based on current application demand. It commonly uses CloudWatch metrics to trigger scaling actions.
Simple ga gurthupettuko:
Dynamic Scaling ante current demand batti EC2 instances ni automatic ga scale cheyyadam.
Eng: Dynamic Scaling adjusts instances based on current application demand.

6.What is Target Tracking Scaling?
Target Tracking Scaling automatically adjusts the number of EC2 instances to maintain a specific target value for a metric. For example, we can maintain average CPU utilization around a target value.
Simple ga gurthupettuko:
Target Tracking ante oka specific metric target value ni maintain cheyyadaniki instances ni scale cheyyadam.
Eng: Target Tracking Scaling maintains a target value for a selected metric.

7.What is Step Scaling?
Step Scaling adjusts the number of EC2 instances based on the size of the CloudWatch alarm breach. Different levels of scaling can be configured for different metric ranges.
Simple ga gurthupettuko:
Step Scaling ante metric entha increase or decrease ayindo batti different number of instances add or remove cheyyadam.
Eng: Step Scaling adjusts instances based on the size of a metric change.

8.What is Scheduled Scaling?
Scheduled Scaling changes the desired capacity of an Auto Scaling Group at a specific time. It is useful when application demand follows a predictable schedule.
Simple ga gurthupettuko:
Scheduled Scaling ante specific time ki instances ni increase or decrease cheyyadam.
Eng: Scheduled Scaling changes Auto Scaling capacity at a scheduled time.

9.What is Predictive Scaling?
Predictive Scaling uses historical and forecasted usage patterns to automatically plan scaling actions before expected demand occurs. It is useful for applications with predictable traffic patterns.

Simple ga gurthupettuko:
Predictive Scaling ante future demand ni predict chesi mundugane instances ni scale cheyyadam.
Eng: Predictive Scaling uses forecasted demand to plan scaling actions.

10.What is Instance Refresh?
Instance Refresh is used to gradually replace instances in an Auto Scaling Group with new instances. It is useful when updating the AMI or launch template configuration.
Simple ga gurthupettuko:
Instance Refresh ante old EC2 instances ni gradual ga new configuration tho replace cheyyadam.
Eng: Instance Refresh gradually replaces instances with new configurations.

11.What are Lifecycle Hooks?
Lifecycle Hooks allow you to pause an EC2 instance during launch or termination so that additional actions can be performed. They are useful for tasks such as configuration, log collection, or cleanup.
Simple ga gurthupettuko:
Lifecycle Hook ante EC2 launch or terminate ayye time lo additional tasks cheyyadaniki pause chese feature.
Eng: Lifecycle Hooks pause instances during launch or termination for additional actions.

12.Why do we use Auto Scaling?
We use Auto Scaling to automatically maintain the required number of EC2 instances based on application demand. It helps improve availability, scalability, and cost efficiency.
Simple ga gurthupettuko:
Application demand batti servers automatic ga increase or decrease cheyyadaniki Auto Scaling use chestham.
Eng: We use Auto Scaling to maintain the required number of instances based on demand.

13.What happens if an EC2 instance in an Auto Scaling Group becomes unhealthy?
Auto Scaling detects the unhealthy instance through health checks and terminates it when required. It then launches a replacement instance to maintain the desired capacity.
Simple ga gurthupettuko:
Oka EC2 unhealthy ayite Auto Scaling danini replace chesi new instance launch chestundi.
Eng: Auto Scaling replaces unhealthy instances to maintain the desired capacity.

14.What is Minimum Capacity?
Minimum Capacity defines the minimum number of EC2 instances that an Auto Scaling Group should maintain. Auto Scaling normally does not reduce the group below this value.
Simple ga gurthupettuko:
Minimum Capacity ante Auto Scaling Group lo minimum ga maintain cheyyalsina EC2 instances count.
Eng: Minimum Capacity defines the minimum number of instances in an Auto Scaling Group.

15.What is Maximum Capacity?
Maximum Capacity defines the maximum number of EC2 instances that an Auto Scaling Group can launch. Auto Scaling will not normally increase the group beyond this value.
Simple ga gurthupettuko:
Maximum Capacity ante Auto Scaling Group lo maximum ga create cheyyagalige EC2 instances count.
Eng: Maximum Capacity defines the maximum number of instances in an Auto Scaling Group.

16.What is Desired Capacity?
Desired Capacity is the number of EC2 instances that the Auto Scaling Group tries to maintain. Auto Scaling launches or terminates instances to reach the desired capacity when required.
Simple ga gurthupettuko:
Desired Capacity ante Auto Scaling Group maintain cheyyali anukune EC2 instances count.
Eng: Desired Capacity is the number of instances the Auto Scaling Group tries to maintain.

17.What is Scale Out?
Scale Out means increasing the number of EC2 instances when application demand increases. It helps the application handle additional traffic or workload.
Simple ga gurthupettuko:
Scale Out ante demand perigina appudu EC2 instances ni increase cheyyadam.
Eng: Scale Out increases the number of EC2 instances when demand increases.

18.What is Scale In?
Scale In means decreasing the number of EC2 instances when application demand decreases. It can help reduce unnecessary infrastructure costs.
Simple ga gurthupettuko:
Scale In ante demand taggina appudu EC2 instances ni decrease cheyyadam.
Eng: Scale In decreases the number of EC2 instances when demand decreases.

19.What is the difference between Dynamic Scaling and Scheduled Scaling?
Dynamic Scaling reacts to current application demand or metrics, while Scheduled Scaling changes capacity at predefined times. Dynamic Scaling is useful for changing demand, while Scheduled Scaling is useful for predictable traffic patterns.
Simple ga gurthupettuko:
Dynamic Scaling current demand batti work chestundi, Scheduled Scaling fixed time batti work chestundi.
Eng: Dynamic Scaling reacts to demand, while Scheduled Scaling works at predefined times.

20.What is the difference between Target Tracking and Step Scaling?
Target Tracking tries to maintain a specific target value for a metric, while Step Scaling changes capacity by predefined amounts based on the size of a metric breach.
Simple ga gurthupettuko:
Target Tracking target value maintain chestundi, Step Scaling metric breach batti fixed steps lo scale chestundi.
Eng: Target Tracking maintains a target value, while Step Scaling uses predefined scaling steps.

21.Can Auto Scaling work with a Load Balancer?
Yes. An Auto Scaling Group can be associated with a Load Balancer target group. The Load Balancer distributes traffic to healthy instances while Auto Scaling adjusts the number of instances.
Simple ga gurthupettuko:
Load Balancer traffic distribute chestundi, Auto Scaling EC2 count ni manage chestundi.
Eng: A Load Balancer distributes traffic while Auto Scaling manages the number of instances.

22.How does Auto Scaling maintain high availability?
Auto Scaling can launch instances across multiple Availability Zones and replace unhealthy instances automatically. This helps keep the application available even when an instance or Availability Zone has problems.
Simple ga gurthupettuko:
Multiple AZs lo instances run chesi unhealthy servers ni replace cheyyadam valla high availability maintain chestham.
Eng: Auto Scaling improves availability by running and replacing instances across Availability Zones.

23.How would you troubleshoot an Auto Scaling issue?
I would check the Auto Scaling Group desired, minimum, and maximum capacity, launch template, scaling policies, CloudWatch alarms, health checks, instance status, and activity history.
Simple ga gurthupettuko:
Auto Scaling problem vaste ASG settings → Launch Template → Scaling Policy → CloudWatch → Health Check → Activity History check chestham.
Eng: For an Auto Scaling issue, I would check ASG settings, launch templates, policies, alarms, health checks, and activity history.

24.What happens when the desired capacity is higher than the current instances?
Auto Scaling launches new EC2 instances until the number of running instances reaches the desired capacity, as long as the maximum capacity and other configuration settings allow it.
Simple ga gurthupettuko:
Desired Capacity current instances kanna ekkuva unte Auto Scaling new instances launch chestundi.
Eng: Auto Scaling launches instances to reach the desired capacity.

25.What is the most important thing to remember about Auto Scaling?
Auto Scaling automatically adjusts the number of EC2 instances based on application demand and scaling policies. The important concepts are Auto Scaling Groups, Launch Templates, Scaling Policies, Dynamic Scaling, Target Tracking, Step Scaling, Scheduled Scaling, Predictive Scaling, Instance Refresh, and Lifecycle Hooks.
Simple ga gurthupettuko:
Auto Scaling ante demand batti EC2 instances ni automatic ga increase or decrease chese service.
Eng: Auto Scaling automatically adjusts EC2 instances based on demand and scaling policies.